

AI BASED FAKE NEWS DETECTION TOOL

AI/ML Internship Project Report

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I sincerely thank CODEVEDX for providing me the opportunity to complete this internship project. This work strengthened my understanding of Python, Natural Language Processing (NLP), Machine Learning, and text classification through practical implementation.

Abstract

The AI Based Fake News Detection Tool is a console-based Machine Learning application that classifies news articles as REAL or FAKE. The project uses TF-IDF Vectorization for feature extraction and Logistic Regression for classification. It also provides confidence scores and stores the trained model for future predictions.

Objectives

- Develop a Fake News Detection System.
- Perform text preprocessing using NLP.
- Train a Logistic Regression model.
- Predict REAL or FAKE news.
- Display prediction confidence.
- Save the trained model.

Problem Statement

Fake news spreads rapidly across digital platforms. Manual verification is slow and unreliable, making automated detection an important requirement.

Proposed Solution

The system preprocesses news text using TF-IDF Vectorization, trains a Logistic Regression classifier, predicts whether the news is REAL or FAKE, and displays a confidence score.

Software Requirements

Component	Requirement
Operating System	Windows 10 / 11

Language	Python 3.x
IDE	Visual Studio Code
Libraries	Pandas, Scikit-learn, Joblib

Technologies Used

- Python
- Pandas
- Scikit-learn
- TF-IDF Vectorizer
- Logistic Regression
- Joblib
- CSV Dataset

Workflow

Start → Load Dataset → Preprocess Text → TF-IDF Vectorization → Train Model → Save Model → Enter News → Predict REAL/FAKE → Display Confidence → End

Features

- View Dataset
- Dataset Summary
- Train Machine Learning Model
- Detect Fake News
- Prediction Confidence
- Prediction History
- Search News
- Exception Handling

Advantages

- Fast and automatic detection
- Easy-to-use interface
- Reusable trained model
- Confidence score for predictions

Future Scope

- Flask Web Application
- Support multiple languages
- Deep Learning models (BERT/LSTM)
- Live News API integration

Conclusion

The project successfully demonstrates the application of Machine Learning and Natural Language Processing for fake news detection. It provides reliable classification, confidence scores, and a reusable trained model while improving practical AI and Python development skills.

References

- [Python Documentation](#)
- [Pandas Documentation](#)
- [Scikit-learn Documentation](#)
- [Joblib Documentation](#)